Technical Document #748: Retrieval Augmented Generation - Comprehensive Overvi

Technical Document #748: Retrieval Augmented Generation - Comprehensive Overview

Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Evaluation metrics for RAG include faithfulness, answer relevance, and context recall. They measure the quality of the retrieval-generation pipeline.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Key Takeaways:

- Hallucination detection identifies when LLMs generate factually incorrect information.

- Retrieval-Augmented Generation (RAG) combines information retrieval with text generation.

- Hybrid search combines keyword-based and semantic search.